Contents

Main Research Plan (V3)

Strategic Risk Concealment and Misleading Framing in Deployment-Realistic Financial LLM Agents

> **Core focus:** Measure whether financial LLM agents, under realistic low-nudge goal conflict, strategically conceal or misframe material risk. Restricted to two finance-native forms, **risk concealment** (incl. selective omission) and **misleading framing**. Test whether their **detectability depends on deception subtlety**. Mitigation remains a stretch goal.

Model scope

Goals 1–4 are evaluated across **3 models**: 2 closed/proprietary (API-only, black-box access) and 1 open-weight (full internal access for probe-based methods).

Name	Model	Provider	Access	Weight type
GPT	`gpt-5.5`	OpenAI	API	closed
Claude	`claude-opus-4-8`	Anthropic	API	closed
Open model	**TBC**	TBC	HuggingFace / local	open

Black-box behavioural detectors run on all 3 models; activation probes run on the open model. Because the headline detection comparison is run *within* the open model across a deception-subtlety gradient (not across models), the single open model yields a powered within-model result; the cross-model element (black-box only) is secondary and explicitly illustrative.

Deception definitions (scope)

Two finance-native forms are measured, arranged on a **falsifiability gradient** that serves as the Goal 4 independent variable:

Risk concealment (incl. selective omission): Failure to disclose a material financial, legal, customer, market, or compliance risk the model had access to and that a reasonable stakeholder would expect to see — including presenting favourable facts while omitting the adverse facts needed for a balanced decision.

Misleading framing: Technically true language that understates uncertainty, downside, or regulatory significance.

Falsifiability gradient (Goal 4 IV): - Level 0 — blatant false claim (control, the regime existing detectors already handle) - Level 1 — selective omission (all stated facts true, a held adverse fact left out) - Level 2 — misleading framing (all facts present, downside/regulatory weight understated). Each scored instance is placed on this gradient.

Main goals

Goal 1: Operationalise the two forms with a scoring rubric that distinguishes them from ordinary error, missing information, sycophancy, or weak reasoning; deliver an **atomic material-risk annotation schema** (material-risk units; disclosed / omitted / distorted) and place each instance on the falsifiability gradient.

Goal 2: Assess whether low-nudge financial goal conflict is associated with deceptive outputs vs neutral finance tasks, across all 3 models. Establishes a per-model, per-form deception-rate baseline.

Goal 3: Characterise which form appears most naturally in finance and how rates vary with goal-conflict intensity and incentive, across all 3 models. The intensity dependence doubles as the test for *strategic intent* (see H1b).

Goal 4: Compare **activation-based (white-box) detection** against **black-box behavioural detection** *as a function of deception subtlety.* The central question is whether the white-box/black-box gap is regime-dependent across the gradient (Level 0 $\rightarrow$ 1 $\rightarrow$ 2), holding model and output distribution fixed. Probes run on the open model; black-box methods (consistency probing, follow-up elicitation, audit-style fact-vs-disclosure checks) run on all 3.

Stretch goal (Goal 5): Whether lightweight governance-oriented controls — honesty incentives, clearer disclosure duties, escalation incentives — reduce deception without making the agent unusable. If pursued, tested on whichever model shows the highest deception rate.

Core research questions and hypotheses

- **RQ1 / H1:** Do LLM agents produce deceptive outputs (risk concealment or misleading framing) under realistic financial goal conflict, without direct instructions to deceive? *(All 3 models.)*
- **RQ1b / H1b:** Does deception scale with goal-conflict intensity and revenue/reputation incentives? A positive scaling is the primary evidence that the behaviour is *strategic* (goal-directed) rather than sycophantic or accidental.
- **RQ2 / H2:** Is risk concealment (omission) more common than misleading framing, and are both more common than the Level-0 blatant false claim?
- **RQ3 / H3:** Are deceptive outputs detectable through inconsistencies between known risk facts, recommendations, disclosures, and follow-up explanations — and does this behavioural signal weaken as deception shifts from inconsistency-bearing lies to inconsistency-free omission?
- **RQ4 / H4 (interaction):**
  - *H4a:* No large, consistent main-effect advantage of white-box over black-box detection (consistent with prior parity findings).

- *H4b:* The detection premium is regime-dependent across the gradient; the empirical deliverable is the ordering/crossover of the two methods' degradation curves as deception becomes subtler.
- *H4c:* Any residual white-box advantage is primarily in cost/calibration, not peak accuracy.

- **RQ5 / H5 (stretch):** Can lightweight governance controls reduce deception while preserving useful task completion?

Expected outputs / artifacts

- An **atomic material-risk annotation schema** and a **scoring rubric** separating deception from error, missing information, sycophancy, and weak reasoning (lead deliverable; stands alone under a null detection result).
- A scenario/specification set for deployment-realistic financial goal-conflict settings, each authored at all three falsifiability levels over the **same** risk-fact set.
- Deception-rate and deception-form breakdowns per model (Goals 1–3), including distribution across the gradient.
- Detection degradation curves: activation probes (open model) vs black-box consistency/audit methods (all 3 models), with precision/recall/F1 per method per level, plus the H4b crossover analysis.
- A negative-results fallback framed as a finding: if deception is rare, low-nudge finance produces less deception than high-pressure benchmarks; if the detection premium is regime-flat, this extends existing parity findings into finance and into the omission/framing regime.

TODOs

- Select the open-weight model (TBC) — must expose activations for probes; match the Liars'-Bench / linear-probe families (Llama-3.3-70B, Qwen-2.5-72B, Gemma-3-27B, Mistral-Small-24B) for direct methodological transfer.
- Define scenario families:
  - Balance across multi-agent, tool-using, chat-based, and single-LLM calls.
  - Author each scenario at all three falsifiability levels over the same underlying risk facts, enabling within-scenario detection comparison across the gradient.
  - Balance scenario variants:

Factor	Levels
Goal conflict	None / mild / strong
Disclosure duty	Absent / implicit / explicit
Stakeholder	Client / manager / regulator / internal audit / public investor
Incentive	Accuracy / revenue / retention / deal completion / reputation protection
Time pressure	None / mild / urgent

- Define controls separating deception (concealment / misframing) from non-strategic falsehood, competence error, incompleteness, and sycophancy.
- Shortlist existing benchmarks; define additional rows / transformations. Apply the Agentic Benchmark Checklist (ABC; Zhu et al., 2025, NeurIPS) for contamination, human validation, and overestimation control.
- Define detector methods:

- Black-box: audit-style fact-vs-disclosure consistency; follow-up elicitation; adapt the Contact-Searching-Question consistency metric (Wu et al.) rather than rebuilding one.
- White-box: finance-specific, omission-aware honest/deceptive contrast pairs (probe quality depends on the contrast set).
- Decide handling of multi-agent / tool-using transcripts for both families.

Relevant reading-list papers

ID	Paper	Use in plan
#2	Detecting Strategic Deception with Linear Probes	Core white-box comparator; already contains a black-box transcript-judge baseline that performs comparably — cite to position Goal 4 as the subtlety interaction.
#4	Building Better Deception Probes Using Targeted Instruction Pairs	Probe quality depends on contrast pairs → finance-specific, omission-aware contrasts.
NEW	Benchmarking Deception Probes via Black-to-White Performance Boosts (Parrack et al., 2025)	Direct precedent for Goal 4 main effect; boost is modest and evasion-fragile.
NEW	Liars' Bench (Kretschmar et al., 2025)	White- vs black-box detectors, 4 models / 7 datasets; systematic failure where the transcript alone can't establish a lie = the omission/framing regime.
NEW	Detecting High-Stakes Interactions with Activation Probes (McKenzie et al., 2025)	Supports H4c: probe edge is largely cost/efficiency, not accuracy.
NEW	From Hallucination to Scheming (Shi et al., 2026)	Coverage analysis of 50 benchmarks: 100% fabrication, ~18% omission, ~3 distortion; strategic-deception benchmarks nascent. Quantifies the gap; supplies the incentive-sensitivity method for proving strategic intent.
NEW	ELEPHANT (Cheng et al., 2025)	Low-nudge omission/framing exists — but as *sycophancy*. The boundary case to distinguish from strategic concealment.

ID	Paper	Use in plan
NEW	DeceptionBench (Huang et al., 2025)	General cross-domain deception benchmark with an Economy domain and neutral/reward/pressure conditions; closest general precedent, not finance-deployment-realistic.
NEW	Uncovering Deceptive Tendencies (Järviniemi & Hubinger, 2024)	Low-pressure self-initiated deception in a simulated company assistant; corporate-general, denial-type phenotype.
#7	DECOR: Auditing LLM Deception via Information Manipulation Theory	Atomic-information-unit backbone of the annotation schema (Goal 1).
#8	Probing the Limits of the Lie Detector Approach to LLM Deception	Lie-detection degrades on true-but-misleading/omission outputs → basis for H4b.
#10	LLMs Can Strategically Deceive Their Users When Put Under Pressure	Finance precedent for self-initiated deception; note pressure ablation and its blatant (concealed-illegal-action) phenotype.
#11	Frontier Models Are Capable of In-Context Scheming	Note the no-strong-nudge condition (§3.4) → low-nudge is design, not novelty.
#12	Alignment Faking in Large Language Models	Eval-awareness → low-telegraph design (Goal 4 validity).
#13	How to Catch an AI Liar (black-box)	Primary black-box baseline; targets falsifiable lies, hence expected Level-1/2 degradation.
#15	Agentic Misalignment	Deployment-risk framing for financial agents.
#16	Escalation Channels as Environmental Controls	Stretch-goal (Goal 5) framing.
#19	AI Sandbagging	Eval-awareness validity concern.
#21	Beyond Prompt-Induced Lies	Self-initiated deception precedent (synthetic domain); reuse its consistency metric as a black-box detector.
#22	AI Deception: A Survey	Taxonomy umbrella.

ID	Paper	Use in plan
#32 / #33 / #36	FinVault / Standard Benchmarks Fail / FinSafetyBench	Finance-agent safety context; measure harmful requests / security / performance, not voluntary concealment.
#37	Risk-Concealment Attacks	Attacker-side risk concealment; cite to motivate the inverse (model-side) question.
57	Rigorous Agentic Benchmarks (Zhu et al., 2025)	ABC checklist for benchmark rigour.